Class 8: Mini-Project

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```r
fna.data <- read.csv("WisconsinCancer.csv")

wisc.df <- data.frame(fna.data, row.names=1)
head(wisc.df, 4)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
         perimeter_worst area_worst smoothness_worst compactness_worst
```

```
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
```

```
wisc.data <- wisc.df[,-1]
```

```
diagnosis <- wisc.df$diagnosis
```

Q1. How many observations are in this dataset?

```
dim(wisc.data)
```

```
[1] 569  30
```

There are 569 observations of 30 variables - not including the id variable for wisc.data. There are 569 observations for the diagnosis vector.

Q2. How many of the observations have a malignant diagnosis?

```
sum(grepl("M", diagnosis))
```

```
[1] 212
```

Out of the 569 diagnoses 212 of them are malignant diagnoses.

Q3. How many variables/features in the data are suffixed with _mean?

```
column_names = colnames(wisc.data)

sum(grepl("_mean", column_names))
```

```
[1] 10
```

Out of the 30 variables, 10 of the variables are suffixed with _mean.

```r
colMeans(wisc.data)
```

```
            radius_mean           texture_mean         perimeter_mean 
           1.412729e+01           1.928965e+01           9.196903e+01 
              area_mean        smoothness_mean       compactness_mean 
           6.548891e+02           9.636028e-02           1.043410e-01 
         concavity_mean    concave.points_mean          symmetry_mean 
           8.879932e-02           4.891915e-02           1.811619e-01 
 fractal_dimension_mean              radius_se             texture_se 
           6.279761e-02           4.051721e-01           1.216853e+00 
           perimeter_se                area_se          smoothness_se 
           2.866059e+00           4.033708e+01           7.040979e-03 
         compactness_se           concavity_se      concave.points_se 
           2.547814e-02           3.189372e-02           1.179614e-02 
            symmetry_se   fractal_dimension_se            radius_worst 
           2.054230e-02           3.794904e-03           1.626919e+01 
          texture_worst        perimeter_worst              area_worst 
           2.567722e+01           1.072612e+02           8.805831e+02 
       smoothness_worst      compactness_worst        concavity_worst 
           1.323686e-01           2.542650e-01           2.721885e-01 
   concave.points_worst         symmetry_worst fractal_dimension_worst 
           1.146062e-01           2.900756e-01           8.394582e-02 
```

```r
apply(wisc.data,2,sd)
```

```
            radius_mean           texture_mean         perimeter_mean 
           3.524049e+00           4.301036e+00           2.429898e+01 
              area_mean        smoothness_mean       compactness_mean 
           3.519141e+02           1.406413e-02           5.281276e-02 
         concavity_mean    concave.points_mean          symmetry_mean 
           7.971981e-02           3.880284e-02           2.741428e-02 
 fractal_dimension_mean              radius_se             texture_se 
           7.060363e-03           2.773127e-01           5.516484e-01 
           perimeter_se                area_se          smoothness_se 
           2.021855e+00           4.549101e+01           3.002518e-03 
         compactness_se           concavity_se      concave.points_se 
           1.790818e-02           3.018606e-02           6.170285e-03 
```

```
           symmetry_se    fractal_dimension_se            radius_worst
          8.266372e-03            2.646071e-03            4.833242e+00
         texture_worst         perimeter_worst              area_worst
          6.146258e+00            3.360254e+01            5.693570e+02
      smoothness_worst       compactness_worst         concavity_worst
          2.283243e-02            1.573365e-01            2.086243e-01
  concave.points_worst          symmetry_worst fractal_dimension_worst
          6.573234e-02            6.186747e-02            1.806127e-02
```

```r
wisc.pr <- prcomp(wisc.data, scale = TRUE)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

PC1: proportion of variance = 0.4427. 44.27% of the original variance is captured by PC1.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

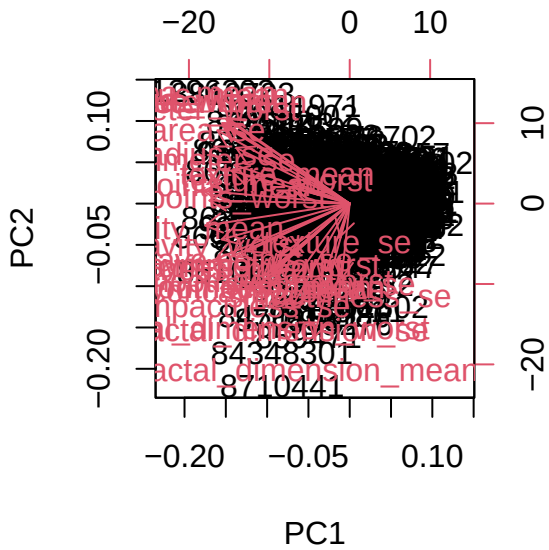
PC3: cumulative proportion = 0.72636. It takes 3 principal components to describe at least 70% of the original variance in the data.

> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

PC7: cumulative proportion = 0.91010. It takes 7 principal components to describe at least 90% of the original variance in the data.

> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

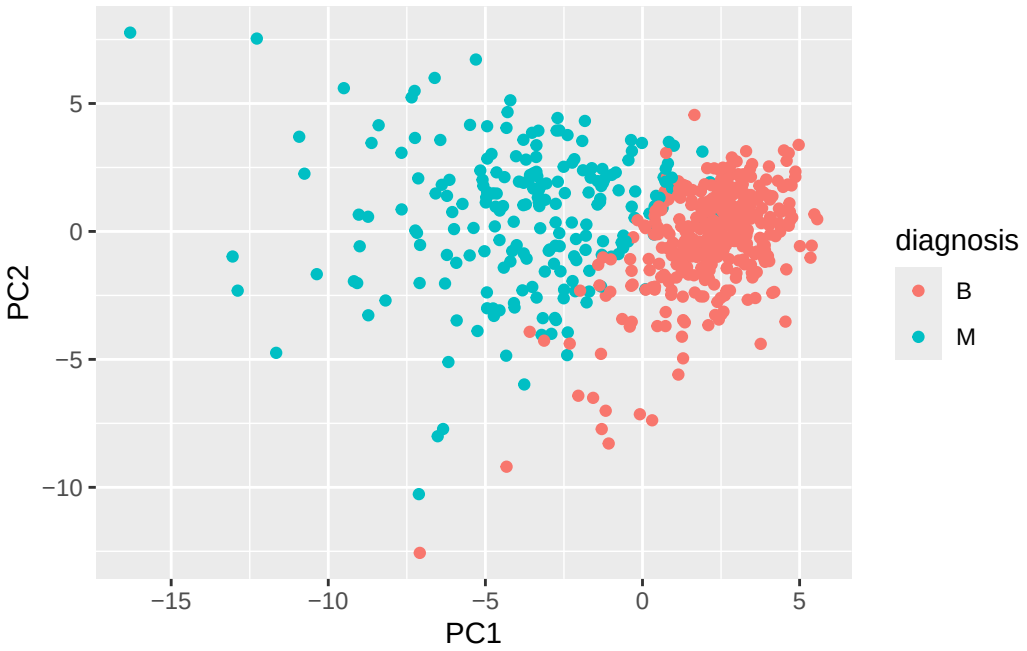
```
biplot(wisc.pr)
```



It is absolutely unreadable. I can't read or distinguish any data points.

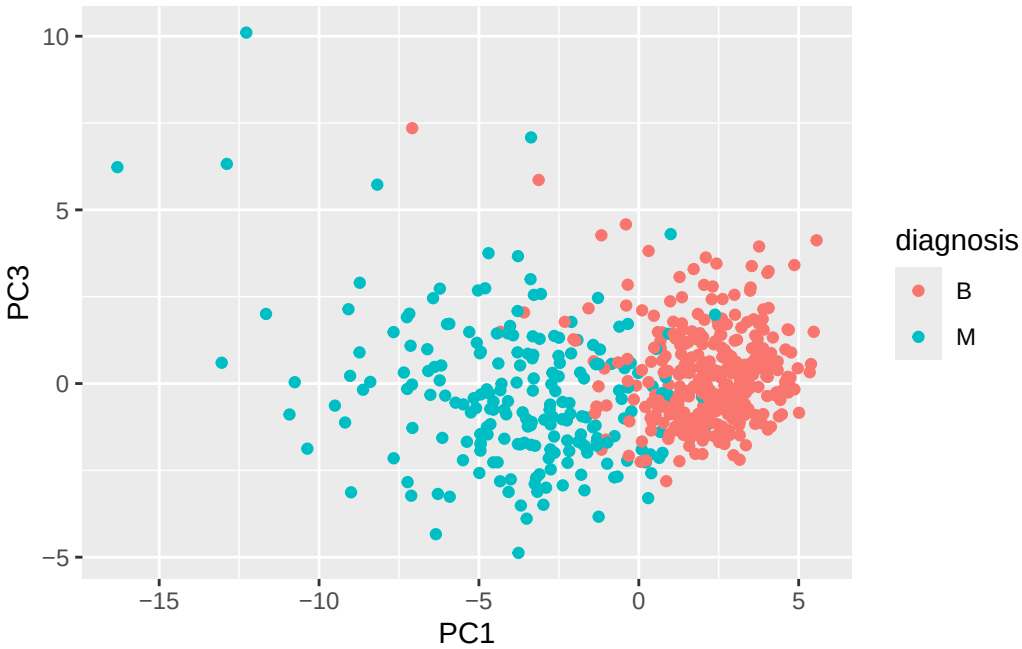
```
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(x = PC1, y = PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



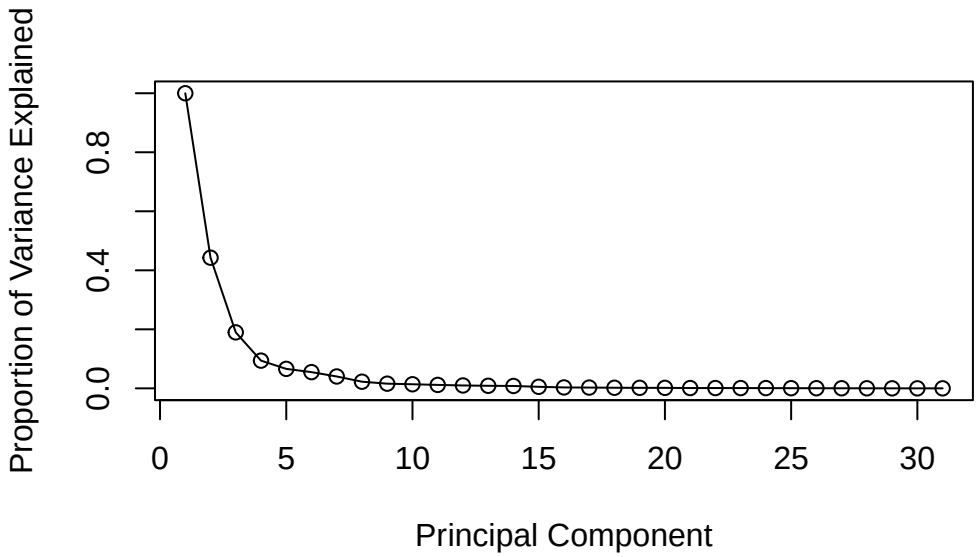
Both plots are centered at 0,0, however, PC1 vs PC3 has a tighter grouping and less outliers. PC1 is capturing the separation of malignant from benigned samples.

```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

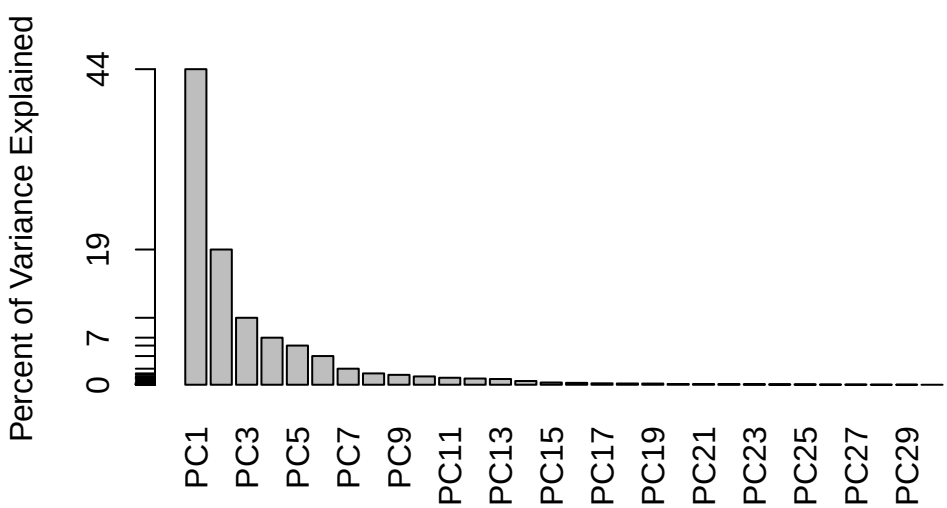
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
pve <- pr.var / sum(pr.var)

plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```




```
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation[,1]
```

```
            radius_mean           texture_mean         perimeter_mean 
            -0.21890244            -0.10372458            -0.22753729 
              area_mean        smoothness_mean       compactness_mean 
            -0.22099499            -0.14258969            -0.23928535 
         concavity_mean    concave.points_mean          symmetry_mean 
            -0.25840048            -0.26085376            -0.13816696 
 fractal_dimension_mean              radius_se             texture_se 
            -0.06436335            -0.20597878            -0.01742803 
           perimeter_se                area_se          smoothness_se 
            -0.21132592            -0.20286964            -0.01453145 
         compactness_se           concavity_se      concave.points_se 
            -0.17039345            -0.15358979            -0.18341740 
            symmetry_se   fractal_dimension_se            radius_worst 
            -0.04249842            -0.10256832            -0.22799663 
          texture_worst        perimeter_worst             area_worst 
            -0.10446933            -0.23663968            -0.22487053 
       smoothness_worst      compactness_worst        concavity_worst 
            -0.12795256            -0.21009588            -0.22876753 
   concave.points_worst         symmetry_worst fractal_dimension_worst 
            -0.25088597            -0.12290456            -0.13178394 
```

```
sort(abs(wisc.pr$rotation[,1]), decreasing = TRUE)
```

```
    concave.points_mean          concavity_mean    concave.points_worst 
             0.26085376              0.25840048              0.25088597 
       compactness_mean         perimeter_worst         concavity_worst 
             0.23928535              0.23663968              0.22876753 
            radius_worst          perimeter_mean              area_worst 
             0.22799663              0.22753729              0.22487053 
              area_mean             radius_mean            perimeter_se 
             0.22099499              0.21890244              0.21132592 
      compactness_worst               radius_se                 area_se 
             0.21009588              0.20597878              0.20286964 
       concave.points_se          compactness_se            concavity_se 
```

```
       0.18341740              0.17039345              0.15358979
 smoothness_mean           symmetry_mean fractal_dimension_worst
       0.14258969              0.13816696              0.13178394
smoothness_worst          symmetry_worst           texture_worst
       0.12795256              0.12290456              0.10446933
    texture_mean    fractal_dimension_se  fractal_dimension_mean
       0.10372458              0.10256832              0.06436335
     symmetry_se               texture_se           smoothness_se
       0.04249842              0.01742803              0.01453145
```

-0.26085376. Nope, concave.points_mean has the largest contribution.

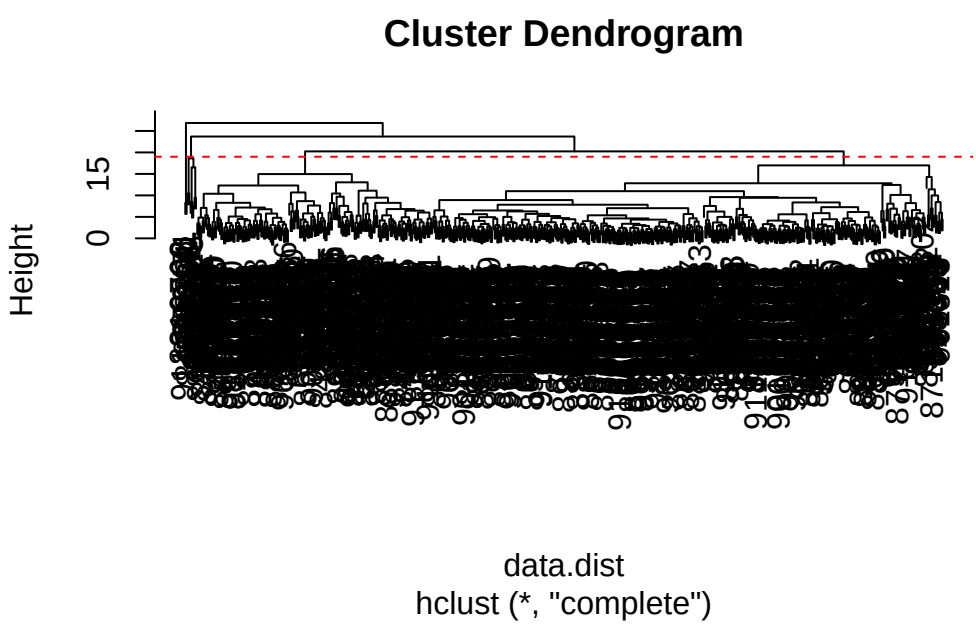
```r
data.scaled <- scale(wisc.data)
```

```r
data.dist <- dist(data.scaled)
```

```r
wisc.hclust <- hclust(data.dist, method = "complete")
```

Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

```r
plot(wisc.hclust)
abline(h = 19, col="red", lty=2)
```



At height = 19, there are 4 clusters.

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

```
wisc.c <- hclust(data.dist, method = "complete")
wisc.s <- hclust(data.dist, method = "single")
wisc.a <- hclust(data.dist, method = "average")
wisc.w <- hclust(data.dist, method = "ward.D2")

table(cutree(wisc.c, k= 4), diagnosis)
```

```
   diagnosis
      B   M
  1  12 165
  2   2   5
  3 343  40
  4   0   2
```

```
table(cutree(wisc.s, k= 4), diagnosis)
```

```
   diagnosis
      B   M
  1 356 209
  2   1   0
  3   0   2
  4   0   1
```

```
table(cutree(wisc.a, k= 4), diagnosis)
```

```
   diagnosis
      B   M
  1 355 209
  2   2   0
  3   0   1
  4   0   2
```

```
table(cutree(wisc.w, k= 4), diagnosis)
```

```
   diagnosis
      B   M
  1   0 115
  2   6  48
  3 337  48
  4  14   1
```

Ward.D2 gives the cleanest separation between the samples. Each row is obvious which B or M is the majority, unlike the others.

```
wisc.pr.hclust <- hclust(dist(wisc.pr$x[,1:7]), method = "ward.D2")
```

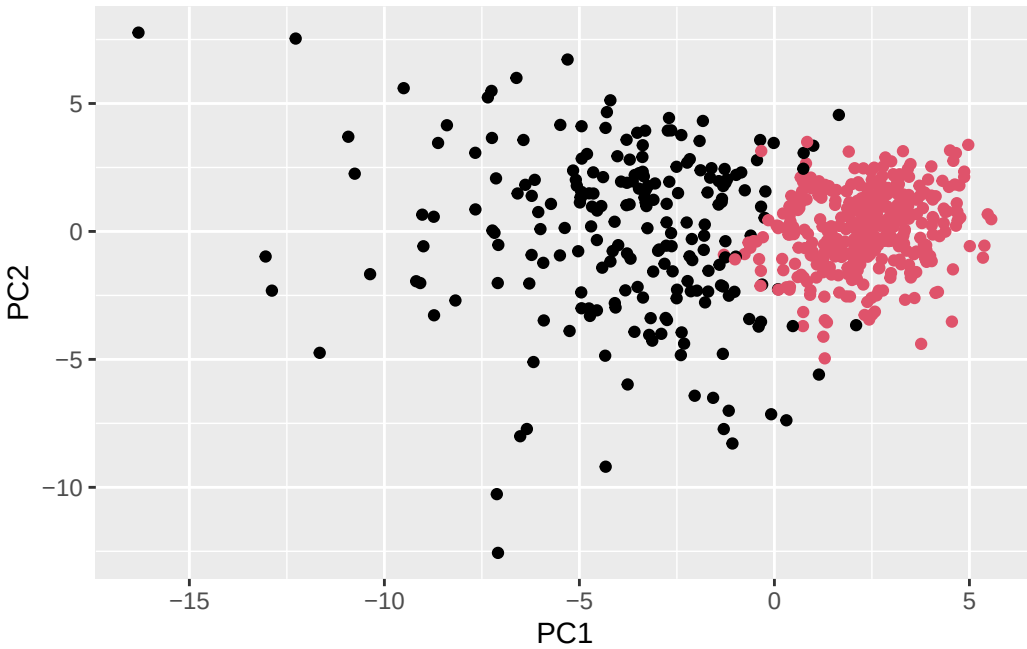
```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1   2
216 353
```

```
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  28 188
   2 329  24
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```




```r
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
```

```r
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

```r
table(wisc.hclust.clusters, diagnosis)
```

```
                     diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

It does a pretty decent job, where out of 216, 188 was identified as malignant. And, out of 353, 329 was identified as benign.

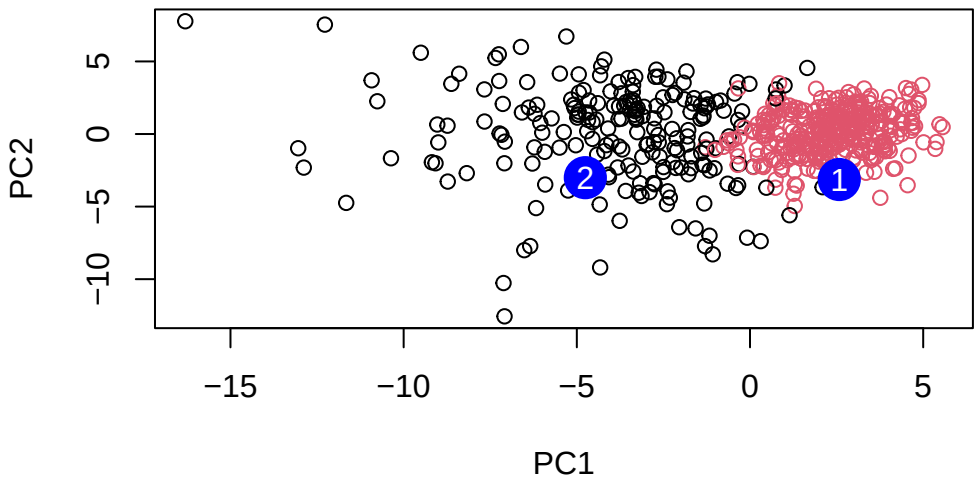
> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

It does comparably worse than the 2 cluster model. For cluster 3 it had 343 B and 40 M which is worse than the 2 cluster model. It also needed 4 cluster for clear separation. Doing PCA first shows substantive improvement.

```
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Patient 2 should be prioritized first because it is on the cluster that was overwhelmingly malignant is the previous table() answer.